**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**
**SCHOOL OF COMPUTER SCIENCE & ENGINEERING**

# Even Semester Examinations 2023-24

**Paper Code: PCC-CS601**

**Paper Name: Compiler Design**

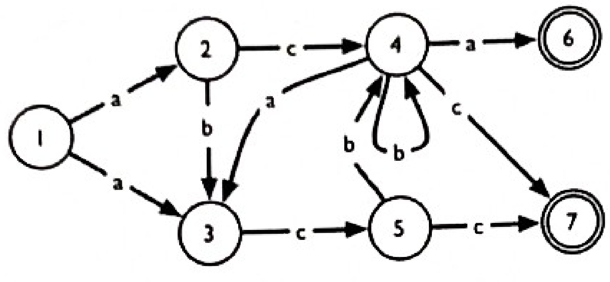
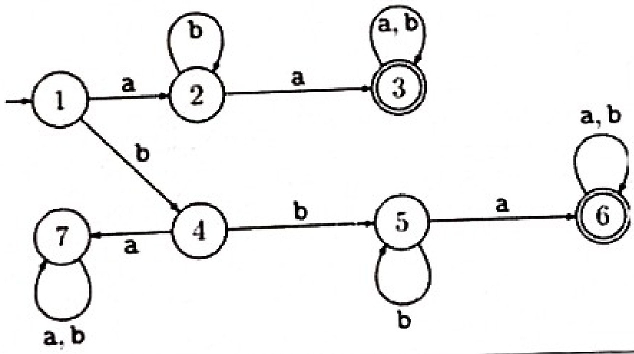
*Time Allotted* :3 Hours *Full Marks:* 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.
**Answer all the questions**

| Sl. | Questions | Marks | Mapped CO | Bloom's Level |
|---|---|---|---|---|
| | **Group A ( 10 x 1 = 10 Marks )**<br>**Answer all questions. Each question carries ONE mark.** | | | |
| 1.(A) | An NFA cannot be used to recognize whether a given input string matches regular expression or not. Hence, the NFA is converted to DFA in a lexical analyzer generator.<br>a) True<br>b) False | 1 | CO1 | BL-1 |
| 1.(B) | Firstpos( ) of a dot(.) node with left child c1 and right child c2 is<br>*a) Firstpos(c1) UNION Firstpos(c2)*<br>*b) Firstpos(c1) INTERSECTION Firstpos(c2)*<br>*c) if (nullable(c1))*<br>*Firstpos(c1) UNION Firstpos(c2)*<br>*else*<br>*Firstpos(c1)*<br>d) *None* | 1 | CO2 | BL-1 |
| 1.(C) | The following code segment can be optimized in _________ time by _________ method.<br>*Pi := 3.14;*<br>*Area := Pi * r ** 2;*<br>a) run, constant folding<br>b) compile, constant folding<br>c) run, constant propagation<br>d) compile, constant propagation | 1 | CO5 | BL-3 |
| 1.(D) | Which of the following symbol table implementation is best suited if access time is to be minimum used?<br>a) Search tree<br>b) Linear list<br>c) Hash table<br>d) Binary search tree | 1 | CO1 | BL-2 |
| 1.(E) | An ideal compiler should<br>a) be smaller in size<br>b) take less time for compilation<br>c) be written in a high-level language<br>d) produce object code that is smaller in size and execute faster | 1 | CO1 | BL-1 |

| | | | | |
|---|---|---|---|---|
| 1.(F) | A bottom-up parser generates<br>a) left-most derivation<br>b) right-most derivation<br>c) left-most derivation in reverse<br>✓d) right-most derivation in reverse | 1 | CO1 | BL-1 |
| 1.(G) | Reduction in strength means<br>a) removing common sub-expression<br>b) removing loop invariant computation<br>✓c) replacing a costly operation by a relatively cheaper one<br>d) replacing run time computation by compile time computation | 1 | CO1 | BL-1 |
| 1.(H) | Which is/are operator grammar in the given set?<br>a) $E \rightarrow AB$<br>$A \rightarrow a$<br>$B \rightarrow b$<br>✓b) $E \rightarrow E + E \mid E * E \mid id$<br>c) $E \rightarrow E + E \mid E * E \mid id \mid \epsilon$<br>d) None | 1 | 3 | BL-3 |
| 1.(I) | Consider the following grammar G = (V, T, P, S} where, V = {S, C, $S_1$}, T = {a, b, i, t, e} and P = {$S \rightarrow iCtSS_1 \mid a$, $S_1 \rightarrow eS \mid \epsilon$, $C \rightarrow b$}<br>The grammar is not LL(1) because<br>a) it is left recursive<br>✓b) it is ambiguous<br>c) it is right recursive<br>d) it is not context free | 1 | CO3 | BL-3 |
| 1.(J) | Type checking is normally done during<br>a) parsing<br>b) syntax analysis<br>✓c) semantic analysis<br>d) none | 1 | 1 | BL-1 |
| | **Group B ( 3 x 5 = 15 Marks)**<br>**Answer all questions. Each question carries FIVE marks.** | | | |
| 2.(A) | a) Write the lexemes, generated by the lexical analyzer for the following source code.<br>a = ++b + c++;<br>/* this is for addition */<br>for ( i =0; i < 5.12; i++);<br>// /* for loop | 3 | CO1 | BL-3 |
| | b) Find the regular expression that generates strings like "*bacterious*" and "*tragedious*" that contain all five vowels, in order, but appearing only once each. | 2 | CO2 | BL-3 |
| OR 2.(B) | a) Show the compiler output and interpreter output for the following **C** source code segment.<br>a = 5;<br>b = 3;<br>sum = a * 2 + b; | 3 | CO1 | BL-3 |
| | b) Write a ~~CGF~~ CFG for the language of the regular expression $(ab)^*(ba)^*$. | 2 | CO2 | BL-3 |
| 3.(A) | Define synthesized and inherited attribute. Explain with examples, the application of these attributes in C language. | 5 | CO4 | BL-2 |
| OR 3.(B) | Write semantic action for bottom-up evaluation of the following grammar:<br>$A \rightarrow ( A ) \mid -F \mid +F \mid A + A \mid num$ | 5 | CO4 | BL-3 |
| 4.(A) | Describe the issues to be addressed during code generation. | 5 | CO5 | BL-1 |
| OR 4.(B) | Write three address code (TAC) for the assignment statement: x = (a + b) + (a – c) + (a – c).<br>Construct the DAG for the TAC. | 5 | CO5 | BL-3 |

**Group C ( 3 x 15 = 45 Marks)**
**Answer all questions. Each question carries FIFTEEN marks.**

| | | | | |
|---|---|---|---|---|
| 5.(A) | a) Explain the phases of a compiler with a neat diagram. Show the output of each phase of the compiler for the expression: a = b + c * 4.5 (Here, a, b and c are of type integer.) | 8 | CO1 | BL-3 |
| | b) Convert the following NFA into an equivalent DFA. Show each step clearly.  | 7 | CO2 | BL-3 |
| OR 5.(B) | a) Suppose you had to write a program to count the number of if statements used in a C program. Does the obvious approach to count the number of matches with the sub-string "if" produce the same result? Justify your answer. | 3 | CO1 | BL-5 |
| | b) Show the position of a compiler in a language processing system with a neat block diagram describing the functions of each block in the system. | 5 | CO1 | BL-1 |
| | c) Minimize the following DFA. Show each step clearly.  | 7 | CO2 | BL-3 |
| 6.(A) | Consider the following Context Free Grammar<br>S → Aa \| BAb<br>A → BB \| c<br>B → Sd \| e<br>where (a, b, c, d, e) is the set of terminal symbols. | | | |
| | a) Compute FIRST and FOLLOW for each nonterminal. Show your work. | 6 | CO3 | BL-3 |
| | b) Construct LL(1) parsing table. | 6 | CO3 | BL-3 |
| | c) Is the grammar LL(1)? Justify your answer. | 3 | CO3 | BL-5 |
| OR 6.(B) | Consider the following grammar:<br>A → aBa \| bBb \| aCb \| bCa<br>B → c<br>C → c | | | |
| | a) Construct LR(1) parsing table. | 12 | CO3 | BL-3 |
| | b) Is the grammar SLR(1)? Why or why not? | 3 | CO3 | BL-5 |
| 7.(A) | a) Consider the following basic block, in which all variables are integers and ** denotes exponentiation.<br>a := b + c<br>z := a ** 2<br>x := 0 * b<br>y := b + c<br>w := y * y<br>u := x + 3<br>v := u + w | | | |

| | | | | |
|---|---|---|---|---|
| | I. Apply the following optimizations to this basic block in the given order. Show the result of each transformation.<br>i) Algebraic simplification<br>ii) Strength reduction<br>iii) Common sub-expression elimination<br>iv) Copy propagation<br>v) Constant folding | 5 | CO5 | BL-3 |
| | II. Is the resulting code after I.(v) optimal? Justify your answer. | 2 | CO5 | BL-5 |
| | III. What optimizations, in what order, can you apply to optimize the result of I.(v) further? | 3 | CO5 | BL-5 |
| | b) Convert the following C code segment to TAC.<br>int a[30];<br>int b;<br>a[21] = 5;<br>b = a[23]; | 5 | CO5 | BL-3 |
| OR<br>7.(B) | a) Consider the following TAC<br>(leader) i := 0<br>n1 := a * b<br>n2 := a - b<br>(leader) L0: t2 := i * 4<br>t3 := &arr<br>t4 := t3[t2]<br>t5 := n1 * n2<br>if t4 > t5 goto L1<br>(leader) goto L2<br>(leader) L1: i := i + 1<br>goto L0<br>(leader) L2: return i<br>goto L3<br>(leader) L3:<br>Identify leaders.<br>Identify basic blocks.<br>Construct flow-graph. | | | |
| | i) Identify leaders. | 4 | CO5 | BL-3 |
| | ii) Identify blocks. | 3 | CO5 | BL-3 |
| | iii) Construct flow-graph. | 3 | CO5 | BL-3 |
| | b) Construct activation tree and activation record for the following code.<br>program main;<br>procedure p;<br>var q:real;<br>procedure r;<br>var s:integer;<br>begin ... end;<br>begin r; end;<br>procedure t;<br>var d:real;<br>procedure u;<br>var v:integer;<br>begin ... end;<br>begin u; end;<br>begin p; t; p; end | 5 | CO5 | BL-4 |

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL
SCHOOL OF COMPUTER SCIENCE & ENGINEERING

Even Semester Examinations 2023-24

**Paper Code:** *PCC-CS602*

**Paper Name:** *Computer Network*

*Time Allotted* :3 Hours

*Full Marks :* 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

**Answer all the questions**

## GROUP – A( Multiple Choice Type Questions )

| | | 10X1 = 10 | | |
|---|---|---|---|---|
| 1 | **Choose the correct alternatives of the following :** | | | |
| | | MARKS | Mapped CO | Bloom's Level |
| i) | Which of the following technique(s) is to get the maximum synchronizing capability? | 1 | CO1 | BL-1 |
| a. | Pseudo ternary | | | |
| b. | Non return to zero Inverted (NRZI) | | | |
| c. | Manchester | | | |
| d. | Non Return to Zero Level (NRZ-L) | | | |
| ii) | In Cyclic redundancy checking, what is CRC? | 1 | CO2 | BL-2 |
| a. | The divisor | | | |
| b. | The quotient | | | |
| c. | The dividend | | | |
| d. | The remainder | | | |
| iii) | A cable break in a _______ topology stops all transmission. | 1 | CO2 | BL-1 |
| a. | Mesh | | | |
| b. | Star | | | |
| c. | Primary | | | |
| d. | Bus | | | |
| iv) | A router is involved in ____________ layers | 1 | CO1 | BL-2 |
| a. | two | | | |
| b. | three | | | |
| c. | four | | | |
| d. | five | | | |
| v) | The_________ layer is responsible for the delivery of a message from one process to another. | 1 | CO1 | BL-1 |
| a. | physical | | | |
| b. | transport | | | |
| c. | network | | | |
| d. | application | | | |

| | | | | |
|---|---|---|---|---|
| vi) | What is the Size of the UDP Packet Header in the Transport Layer? | 1 | CO4 | BL-1 |
| a. | 8 bytes | | | |
| b. | 12 bytes | | | |
| c. | 16 bytes | | | |
| d. | 20 bytes | | | |
| vii) | Which of the following address belongs class A? | 1 | CO4 | BL-1 |
| a. | 121.12.12.248 | | | |
| b. | 130.12.12.248 | | | |
| c. | 136.12.12.248 | | | |
| d. | 129.12.12.248 | | | |
| viii) | An IPv4 datagram is fragmented into three smaller datagrams. which of the following is true ? | 1 | CO4 | BL-1 |
| a. | The identification field is same for all three datagrams | | | |
| b. | The do not fragment bit is set to 1 for all three datagrams | | | |
| c. | The more fragment bit is set to 0 for all three datagrams | | | |
| d. | The offset field is same for all three datagrams | | | |
| ix) | An IP fragment has arrived with an offset value of 200. How many bytes of data were originally sent by the source before the data in this fragment? | 1 | CO4 | BL-1 |
| a. | 200 | | | |
| b. | 400 | | | |
| c. | 800 | | | |
| d. | 1600 | | | |
| x) | In an Ipv4 packet, the value of HLEN is $(1000)_2$. How many bytes of options are being carried by this packet ? | 1 | CO4 | BL-1 |
| a. | 2 byte | | | |
| b. | 12 byte | | | |
| c. | 22 byte | | | |
| d. | 32 byte | | | |

**GROUP – B( Short Answer Type Questions )** 3X5 = 15

| | | MARKS | Mapped CO | Bloom's Level |
|---|---|---|---|---|
| 2.a. | Explain operation of each layer of TCP/IP protocol | 5 | CO3 | BL-3 |
| | OR | | | |
| 2.b. | Draw Differential Manchester, Manchester, NRZ-L, NRZ-I & RZ of 101101100 | 5 | CO3 | BL-3 |
| 3.a. | Four data channel (Digital ), each transmitting at 4 Mbps, use a satellite channel of 4 MHz. Design an appropriate configuration using FDM | 5 | CO2 | BL-2 |
| | OR | | | |
| 3.b. | Why is CSMA/CD not useful for Wireless LAN protocol? Differentiate between Hub and bridge. | 2+3 | CO2 | BL-2 |
| 4.a. | Explain auto negotiation. Differentiate between CHAP and PAP packets. | 2+3 | CO3 | BL-3 |
| | OR | | | |
| 4.b. | What is sliding window protocol? Explain selective repeat ARQ, lost frame. | 1+4 | CO3 | BL-3 |

## GROUP – C( Long Answer Type Questions )

**Answer the following.** 3 X 15 = 45

| | | MARKS | Mapped CO | Bloom's Level |
|---|---|---|---|---|
| 5.a. | A receiver receives the code 1100110011l, When it uses the Hamming code algorithm, the result is 0101, Which bit is in error. What is Token passing? Explain controlled access. Given a 10 bit sequence 1010011110 and divisor of 1001, What is CRC? check your answer. | 5+2+2+3 +3 | CO2 | BL-3 |
| | OR | | | |
| 5.b. | i) Explain PPP frame format.<br>ii) Name and explain different HDLC frames.<br>iii)Explain bit staffing algorithm.<br>iv) Differentiate between slotted and Pure ALOHA.<br>v)How can you compare CHAP and PAP packets? | 3+3+3+3 +3. | CO2 | BL-4 |
| 6.a. | i)Compare SVC with PVC.<br>ii) Explain the operation of Frame Relay.<br>iii) Why is the control field from HDLC totally dropped from Frame Relay?<br>iv) Explain transparent bridge.<br>v) How flooding can be removed from a bridge explain.<br>vi)How is hub related to repeater?<br>viii) What is the basis of membership of VLAN | 2+5+1+1 +3+2+1 | CO4 | BL-3 |
| | OR | | | |
| 6.b. | Write a short notes on the following (Any 3)<br>i. VLAN<br>ii. ATM<br>iii. Bluetooth<br>iv. SONET<br>v. TDM<br>vi. CDMA | 5+5+5 | CO4 | BL-2 |
| 7.a.i. | An Ipv4 packet has arrived in which the offset value is 150, the value of HLEN is 5 and the value of total length field is 100. What are the number of first byte and last byte? | 3 | CO3 | BL-3 |
| ii. | A network on the internet has a subnet mask of 255.255.240.0. What is the maximum number of hosts it can handle? | 2 | CO3 | BL-3 |
| iii. | If a Company has 100 computers and use class C Ipv4 address, want to create four sub networks each of 25 computers then what is the best possible subnet mask? | 2 | CO3 | BL-3 |
| iv. | Explain the process of building a Link State packet with example. | 4 | CO3 | BL-2 |
| v. | Draw an Ipv4 header diagram mentioning name and size of each field. | 4 ✓ | CO3 | BL-2 |
| | OR | | | |
| 7.b.i. | What is the difference between flow control in a data link layer and flow control in a transport layer? | 2 | CO3 | BL-2 |
| ii. | Explain Leaky Bucket and Token Bucket methods of achieving good quality of service. | 3 | CO3 | BL-2 |
| iii. | Explain the process of three-way handshaking of connection establishment in TCP mentioning value of each relevant field in TCP header. | 3 | CO3 | BL-2 |
| iv. | What is silly window syndrome? Describe the solution for the following two situation<br>a. Syndrome created by the sender b. syndrome created by the receiver | 4 | CO3 | BL-3 |
| v. | Give name of two flags in TCP header that cannot be used simultaneously. | 1 | CO3 | BL-3 |
| vi. | Explain how TCP, which uses the services provided by the unreliable IP, can provide reliable communication ? | 2 | CO3 | BL-3 |

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

# Even Semester Examinations 2023-24

## Paper Code: OEC -CS601

## Paper Name: Soft Skills and Interpersonal Communication

*Time Allotted* :3 Hours

*Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer all the questions

| Sl | Questions | Marks | Mapped CO | Bloom's Level |
|---|---|---|---|---|
| | **Group A( Multiple Choice/Very Short Type Questions ) Any Ten** | | 10 x 1 = 10 | |
| 1 (a) | What are the key points which you need to cover while giving your self-introduction in an interview? | 1 | CO1 | BL-2 |
| 1 (b) | What are the things which you need to keep in mind to answer a question about your career goals in a company where you are applying? | 1 | CO1 | BL-2 |
| 1 (c) | What are the ingredients required to have confidence in a Group Discussion? | 1 | CO1 | BL-2 |
| 1 (d) | What are the pillars of a Group Discussion? | 1 | CO4 | BL-1 |
| 1 (e) | Why is eye contact regarded as a valuable trait in an interpersonal interaction? | 1 | CO4 | BL-2 |
| 1 (f) | Mention two traits which can present you as confident person in any interpersonal interaction? | 1 | CO4 | BL-3 |
| 1 (g) | On which point do you need to have pre-existing knowledge in order to face the salary negotiation round of an interview successfully? | 1 | CO4 | BL-2 |
| 1 (h) | How is the 'Theory of Mind' applicable while appearing for an interview? | 1 | CO2 | BL-3 |
| 1 (i) | Differentiate CV from Resume. | 1 | CO1 | BL-2 |
| 1 (j) | How do you display your confidence in a Cover Letter via your writing? | 1 | CO3 | BL-2 |
| 1 (k) | What is the problem associated with the practice of False Dichotomy? | 1 | CO5 | BL-2 |
| 1 (l) | What is the problem associated with Is-Ought Fallacy? | 1 | CO5 | BL-2 |
| | **Group B( Short Answer Type Questions )** 3 x 5 = 15 | | | |
| 2. a. | How can the premises of an argument be made acceptable? | 5 | CO5 | BL-2 |
| Or b. | What leads to rejection of the premises of an argument? | 5 | CO5 | BL-2 |
| 3. a. | How are the traits of sympathy and empathy related to each other? | 5 | CO4 | BL-2 |
| Or b. | What is the similarity between False Dichotomy and Is-Ought Fallacy? | 5 | CO4 | BL-2 |
| 4. a. | How does empathy serve as an important leadership skill? | 5 | CO4 | BL-2 |
| Or b. | Why is accountability on the personal front important for a leader? | 5 | CO4 | BL-2 |
| | **Group C( Long Answer Type Questions )** 3 x 15=45 | | | |
| 5.A. | **Read the article and answer the question(s) that follow:**<br>"The government's sudden enforcement of the lock down seemed hastily prepared and immediately disadvantaged already vulnerable populations. There has been a mass exodus of migrant workers and concerns are rising about starvation among people who work in the informal economy. Implementing public health measures is difficult in places with overcrowded living conditions and inadequate hygiene and sanitation. Non-COVID-19 health services have been disrupted. Reports suggest that the government's efforts to provide financial support and a measure of food security to ease these pressures will be insufficient to meet demand. But better planning and communication could have helped avert this crisis."<br>-The Lancet, April 25, 2020 (DOI:https://doi.org/10.1016/S0140-6736(20)30938-7)<br><br>(a) In light of the above passage, does the decision by the Government seem justified while curbing the spread of the disease on one hand and heralding | 7+8 | CO3 | BL-4 |

| | | | | |
|---|---|---|---|---|
| | mass economic crisis to the people in the 'informal economy' on the other? Validate your argument.<br>(b) As a leader of the masses, what could the Government have done to avert the economic crisis imposed on the 'informal economy'? Was there a possible deontological solution to their crisis? | | | |
| Or B. | Read the article and answer the question(s) that follow:<br>"Cognitive offloading has become more common with the introduction of new technologies. Do you rely on your phone to store phone numbers you once memorized? Do you use GPS navigation instead of memorizing your driving routes? Then you know the benefits of cognitive offloading. Cognitive offloading transfers routine tasks to algorithms and robots and frees up your busy mind to deal with more important activities.<br>In an upcoming edition of the peer reviewed journal, *Human Performance in Extreme Environments*, I review the unintended consequences of cognitive offloading in industries like aviation and aerospace. Despite its many benefits, cognitive offloading also introduces a new set of problems. When we offload activities, we also offload learning and judgment. In one study, researchers asked a group of subjects to navigate the streets of London using their own judgment. A second group relied on GPS technology as their guide. The GPS group saw significantly less activity in the brain associated with learning and judgment. In the instance of self-driving cars, drivers may see their driving skills degrade over time."<br>- D. Christopher Kayes, April 22, 2022<br>(a) In light of the above passage, do you think offloading learning and judgement is a better option or are we becoming increasingly reliant on technology? Justify your argument properly.<br>(b) Write a paragraph within 250 words justifying the process of 'cognitive offloading' from a Deontological point of view. | 8+7 | CO3 | BL-4 |
| 6. A. | **Read the article and answer the question(s) that follow:**<br>"Hitler served just nine months of his sentence in the Bavarian fortress of Landsberg am Lech. Here he wrote Mein Kampf, defining his political vision. For him, the state was not an economic entity but racial. He declared the superiority of a white Aryan race, with particular vitriol reserved for the Jews he viewed as "parasites". Their elimination, he said, "must necessarily be a bloody process". Mein Kampf outlined the central tenets of a Germany under Nazi control – military expansion, elimination of "impure" races and dictatorial authoritarianism. After its publication in July 1925, the book saw more exposure for Hitler's views."<br>- Excerpt from article published on BBC Teach<br>(a) In light of the above passage, what was/were the leadership quality(ies) that Adolf Hitler lacked?<br>(b) Why are these qualities important for a leader?<br>(c) Does the quality of empathy require one to exercise sympathy too? Explain. | 5+5+5 | CO5 | BL-2 |
| Or B. | Read the Problem carefully and answer the question (s) that follow:<br>As the General Manager of XYZ Co Ltd, you have been assigned a team of ten people and given a task which is to be completed within a deadline of ten days. On the eighth day, one of your team members who was assigned a vital task requests a leave on grounds of bereavement due to the death of a close one. Your company has hitherto endorsed the culture of firing people who are regarded inept. In this scenario, you have but two options at hand:<br>Option A: Hire a new person to get the job done and fire the existing employee.<br>Option B: Seek an extension of the deadline even if it means that you, as team leader have to take the onus.<br>(a) As a team leader, which option would you choose and why?<br>(b) How would you justify your decision before the higher authority in your company? | 8+7 | CO5 | BL-3 |
| 7. A. | Draft a complete job application with CV and Cover-Letter for the position of Junior Engineer with ABC Co Ltd. | 10+5 | CO1 | BL-3 |
| Or B. | With respect to appearing in a Personal Interview, answer the following questions:<br>(a) If asked to enumerate your weaknesses, what strategy will you adapt to draft an answer to this question?<br>(b) In the salary negotiation round, what strategy should you adapt in order to bag the job as well as not start off being underpaid? | 7+8 | CO4 | BL-3 |

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

**SCHOOL OF COMPUTER SCIENCE & ENGINEERING**

**Even Semester Examinations 2023-24**

**Paper Code: PEC-CS602A**

**Paper Name: Computer Graphics**

***Time Allotted :*** **3 Hours** ***Full Marks :*** **70**

The figures in the margin indicate full marks. Candidates are required to give their answers in their own words as far as practicable.

**Answer all the questions**

| *Sl. No.* | *Questions* | *Marks* | *Mapped CO* | *Bloom's Level* |
|---|---|---|---|---|
| **Group A( Multiple Choice Type Questions ) 10 x 1 = 10** | | | | |
| 1. i) | The components of Interactive computer graphics are –<br>i) A monitor<br>ii) Display controller ✓<br>iii) Frame buffer<br>iv) All of the above | 1 | CO1 | BL-1 Remembering |
| ii) | What is a pixel mask?<br>i) a string containing only 0's<br>✓ ii) a string containing only 1's<br>iii) a string containing two 0's<br>iv) a string containing both 1's and 0's | 1 | CO1 | BL-1 Remembering |
| iii) | Two basic technique for producing color display with a CRT are-<br>i) Shadow mask and random scan<br>✓ ii) Beam penetration method and Shadow mask method<br>iii) Random scan and raster scan<br>iv) None of these | 1 | CO2 | BL-2 Understanding |
| iv) | Bresenham circle algorithm uses the approach of-<br>i) Midpoint<br>ii) Point'<br>iii) Line<br>iv) None of these | 1 | CO2 | BL-2 Understanding |

| | | | | |
|---|---|---|---|---|
| v) | To change the position of a circle or ellipse we translate<br>i) Center coordinates<br>ii) Center coordinates and redraw the figure in new location<br>iii) Outline coordinates<br>iv) All of the mentioned | 1 | CO2 | BL-2 Remembering |
| vi) | In Bresenham's line drawing algorithm, if the distances d1 < d2 then decision parameter Pk is______<br>i) Positive<br>ii) Equal<br>iii) Negative<br>iv) Option a or c | 1 | CO2 | BL-2 Understanding |
| vii) | The process of displaying 3D into a 2D display unit is called as -<br>i. Resolution<br>ii. Projection<br>iii. Rasterization<br>iv. Transformation | 1 | CO1 | BL-1 Remembering |
| viii) | In 2D-translation, a point (x, y) can move to the new position (x', y') by using the equation<br>i) x'=x+dx and y'=y+dx<br>ii) x'=x+dx and y'=y+dy<br>iii)x'=x+dy and y'=y+dx<br>iv)x'=x-dx and y'=y-dy | 1 | CO1 | BL-1 Remembering |
| ix) | Which of the following stores the picture information as a charge distribution behind the phosphor-coated screen?<br>i) Direct-view storage tube<br>ii) Flat panel displays<br>iii) 3D viewing device<br>iv) Cathode ray tube | 1 | CO1 | BL-1 Remembering |
| x) | The center region of the screen and the window can be represented as________<br>i) 0000<br>ii) 1111<br>iii) 0110<br>iv) 1001 | 1 | CO2 | BL-2 Understanding |
| | **Group B( Short Answer Type Questions )** | | | **3 x 5 = 15** |
| 2. A) | What is display processor? Explain its functions. | 2+3 | CO1 | BL-2 Understanding |
| OR 2. B) | Distinguish between Raster and Random Scan display. | 5 | CO1 | BL-2 Understanding |
| 3. A) | With a suitable figure, describe the shadow masking techniques in CRT. | 5 | CO2 | BL-1 Remembering |
| OR 3. B) | What are the characteristics of light? Give the transformation matrix for conversion of RGB to YIQ. | 3+2 | CO2 | BL-2 Understanding |
| 4. A) | Write the boundary fill algorithm for filling a polygon using eight connected approach. | 5 | CO3 | BL-2 Understanding |
| | | | | |

| Group C( Long Answer Type Questions ) | | | | 3 x 15=45 |
|---|---|---|---|---|
| 5. A) | i)Explain the Raster Scan Systems.<br>ii)Describe Output primitives briefly. | 10+5 | CO2 | BL-2 Understanding<br>BL-3 Applying |
| OR<br>5. B) | i)Explain Various Applications of Computer Graphics.<br>ii)Explain Pixel Addressing. | 10+5 | CO2 | BL-2 Understanding |
| 6. A) | i) Draw 8-way symmetry of circle and explain.<br>ii)Write down and explain the midpoint circle drawing algorithm.<br>iii) Use mid-point circle drawing algorithm to plot a circle whose radius =20 units and centre at (50,30). | 3+9+3 | CO3 | BL-3 Applying |
| OR<br>6. B) | i) How RGB model represented?<br>ii) Explain in-details the CMY color model.<br>iii)What are subtractive colors? | 3+9+3 | CO3 | BL-1, BL-4<br>Remembering<br>Analysing |
| 7. A) | i)Explain Midpoint Ellipse Generating Algorithm.<br>ii)Describe Flood Fill algorithm.<br>iii)Explain Scan-Line Polygon Fill Algorithm | 5+5+5 | CO4 | BL-3 Applying |
| OR<br>7. B) | Short note (any three)<br>i) DDA Line Drawing<br>ii) LCD Monitor Vs CRT Monitor<br>iii) Interactive Vs Non-interactive graphics<br>iv) Bresenham's ellipse generating algorithm<br>v) Graphics Pipeline | 3 x 5 | CO4 | BL-1, BL-4<br>Remembering<br>Analysing |

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL
SCHOOL OF COMPUTER SCIENCE & ENGINEERING
Even Semester Examinations 2023-24
Paper Code: PEC-CS601B
Paper Name: Distributed Database

*Time Allotted :* 3 Hours *Full Marks :* 70

The figures in the margin indicate full marks. Candidates are required to give their answers in their own words as far as practicable.

**Answer all the questions**

| GROUP – A( Multiple Choice Type Questions ) | | | | |
|---|---|---|---|---|
| **1** | **Choose the correct alternatives of the following :** | **10 X 1 = 10** | | |
| | | **MARKS** | ***Mapped CO*** | ***Bloom's Level*** |
| i) | A distributed database has which of the following advantages over centralized database ? | 1 | CO1 | BL-1 |
| a. | Software cost | | | |
| b. | Software complexity | | | |
| c. | Slow response | | | |
| d. | Modular growth | | | |
| ii) | An autonomous homogenous environment is which of the following ? | 1 | CO1 | BL-1 |
| a. | The same DBMS is at each node and each DBMS works independently. | | | |
| b. | The same DBMS is at each node and a central DBMS coordinates database access. | | | |
| c. | A different DBMS is at each node and each DBMS works independently | | | |
| d. | A different DBMS is at each node and a central DBMS coordinates database access | | | |
| iii) | Location transparency allows for which of the following ? | 1 | CO1 | BL-1 |
| a. | Users to treat the data as if it is at one location | | | |
| b. | Programmers to treat the data as if it is at one location | | | |
| c. | Managers to treat the data as if it is at one location | | | |
| d. | All of the above | | | |
| iv) | Which of the following is true concerning a global transaction? | 1 | CO1 | BL-1 |
| a. | The required data are at one local site and the distributed DBMS routes requests as necessary | | | |
| b. | The required data are located in at least one non-local site and the distributed DBMS routes requests as necessary. | | | |
| c. | The required data are at one local site and the distributed DBMS passes the request to only the local DBMS. | | | |
| d. | The required data are located in at least one non-local site and the distributed DBMS passes the request to only the local DBMS. | | | |
| v) | Storing a separate copy of the database at multiple locations is which of the following ? | 1 | CO1 | BL-1 |
| a. | Data replication | | | |
| b. | Data sharing | | | |

| | | | | |
|---|---|---|---|---|
| c. | Data mirroring | | | |
| d. | Data fragmentation | | | |
| vi) | Which of the following is an objective of data distribution in the distributed database system? | 1 | CO1 | BL-1 |
| a. | Locality of reference | | | |
| b. | Availability and reliability | | | |
| c. | Communication cost | | | |
| d. | All of these | | | |
| vii) | Which transparency is provided if users are unaware about the location of the data objects ? | 1 | CO1 | BL-1 |
| a. | Naming transparency | | | |
| b. | Location transparency | | | |
| c. | Replication transparency | | | |
| d. | Fragmentation transparency | | | |
| viii) | ___ transparency exists when the end user or programmer must specify both the fragment names and their locations. | 1 | CO1 | BL-1 |
| a. | Local mapping | | | |
| b. | Replication | | | |
| c. | Naming | | | |
| d. | None of these | | | |
| ix) | Some of the columns of a relation are at different sites is which of the following? | 1 | CO1 | BL-1 |
| a. | Data replication | | | |
| b. | Horizontal partitioning | | | |
| c. | Vertical partitioning | | | |
| d. | None of these | | | |
| x) | A fragmentation technique where in every tuple of a table is assigned to one or more fragment as a result of fragmentation is called | 1 | CO1 | BL-1 |
| a. | Vertical fragmentation | | | |
| b. | Horizontal fragmentation | | | |
| c. | Hybrid fragmentation | | | |
| d. | None of these | | | |

## GROUP – B( Short Answer Type Questions )

3 X5 = 15

| Answer the following. | | MARKS | Mapped CO | Bloom's Level |
|---|---|---|---|---|
| 2.a. | Define distributed database. List the advantages of it. | 2+3 | CO1 | BL-1 |
| OR 2.b. | Describe the objectives of data distribution. | 5 | CO1 | BL-1 |
| 3.a. | What is Distributed database? How it differs from Centralized Database ? | 5 | CO1 | BL-1 |
| OR 3.b. | What are homogenous and heterogeneous database. | 5 | CO1 | BL-1 |
| 4.a. | Explain canonical form of global query with suitable example. | 5 | CO2 | BL-2 |
| OR 4.b. | Differentiate Operator tree and Optimization graph with suitable example. | 5 | CO2 | BL-2 |

## GROUP – C( Long Answer Type Questions )

3 X15 = 45

| Answer the following. | | MARKS | Mapped CO | Bloom's Level |
|---|---|---|---|---|
| 5.a.i. | Explain the reference architecture of DDB with a diagram. | 10 | CO1 | BL-2 |
| ii. | "Bottom-up approach of distributed database design is applicable for integrating existing databases"; justify your answer. | 5 | CO1 | BL-2 |

| | | | | |
|---|---|---|---|---|
| OR 5.b.i. | What do you mean by fragmented schema? What are the advantages of keeping fragmentation schema independent of allocation schema? | 2 + 3 | CO1 | BL-2 |
| ii. | Write about the completeness, reconstruction and disjointness condition rules for defining fragmentation. | 6 | CO1 | BL-2 |
| iii | What are the advantages and disadvantages of replication? | 4 | CO1 | BL-2 |
| 6.a.i. | Define transaction. Discuss about the different type of agents in distributed transaction. | 1+5 | CO3 | BL-1 |
| ii. | Write about the atomicity of transactions in distributed databases. | 5 | CO3 | BL-2 |
| iii. | What is network partitioning? | 2 | CO3 | BL-1 |
| iv | Define K-resilient system with respect to site failures | 2 | CO3 | BL-1 |
| OR 6.b.i. | Define distributed serializability. How is it ensured in DDBMS? | 2+3 | CO3 | BL-2 |
| ii. | Differentiate between basic TO algorithm and conservative TO algorithm. | 6 | CO3 | BL-2 |
| iii. | How disadvantages of 2PL are overcome in strict 2PL ? | 4 | CO3 | BL-3 |
| 7.a.i. | What is wait for graph? Differentiate between Local wait for graph (LWFG) and Global wait for graph (DWFG). Explain the use of DWFG with example. | 2+3+5 | CO4 | BL-2 |
| ii. | Explain false deadlock with example. | 5 | CO4 | BL-2 |
| OR 7.b.i. | Differentiate between pre-emptive and non-preemptive methods for distributed deadlock prevention. | 5 | CO4 | BL-2 |
| ii. | How multiple copies of data can be handled by 2-phase locking protocol as a concurrency control mechanism in distributed database? | 5 | CO4 | BL-2 |
| iii. | Compare pessimistic and optimistic concurrency control techniques. | 5 | CO4 | BL-2 |